V2 run report: CPU control missed its frozen gate

2026-10-13 · frozen V2 remediation · 640/640 measured answers checked

V2 retained the V1 curves, targets, algorithms, seeds, caps and arm order, and added a current-thread CPU clock around each complete cold arm. The independent checker reconstructed collection, rank, recovery, descent and all scalars on both models. The same-seed rho A/A CPU ratios were 0.827, 0.924, 0.754 and 0.817. The order-1,613 ratio missed the preregistered 0.8–1.25 gate, so V2′s cost comparison is held as incomplete calibration. The source snapshot `runner-v2.rs` reproduced all deterministic V2 rows.

Duplicate rho control across the retained run versions

Each cell is the full-work duplicate-rho / matched-rho ratio. Both copies performed the same counted group operations on the same target and seed. V1 uses elapsed wall time; V2 and V3 use current-thread CPU time. V2/V3′s frozen admissible A/A band is 0.8–1.25.

subgroup r	V1 wall	V2 CPU	V3 CPU
103	420.987×	0.827×	0.810×
439	0.771×	0.924×	0.918×
1,613	2.071×	0.754×	0.819×
6,427	5.690×	0.817×	0.824×

Pink marks a failed or visibly unstable control. Green marks the frozen CPU ratio gate only. The residual 8–19% A/A CPU imbalance limits the interpretation of the smaller source-versus-destination differences. Source: `summary-v1.tsv`, `summary-v2.tsv`, `summary-v3.tsv` and retained raw records beside them.

The raw, summary, checker and replay records are linked in `REPORT.md`. V3′s separately committed remediation balances both rho copies' order.